

Load Interconnection System Impact Study Report

For Queue Project Q16-0622

South Haven 100 kV

May 20, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0622. The Interconnection Customer proposes a 158.6 MW large load (data center) facility interconnecting at the South Haven 100 kV substation in the Indiana area. The request was received on May 27, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 210 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,944,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MURDOCK / AEP MEDFORD 100.0 kV line	MURDOCK 230/1kV Transformer Bank	109.5	131	44.0
NAAVINE / MURDOCK 100.0 kV	LOWER GRANITE / CANAL ST SUBSTATION 100.0 kV	108.2	133	72.4
SOUTH HAVEN SUBSTATION / NAAVINE 100.0 kV Line	CANAL ST / LOWER GRANITE 100.0 kV L	103.1	129	72.4
LOWER GRANITE / CANAL ST 100.0 kV Line	NAAVINE / MURDOCK 100.0 kV	102.1	160	72.9
FEDERALGEN TAP / AEP MEDFORD SUBSTATION 100.0 kV Line	MURDOCK 230/1 kV Transformer	100.2	223	41.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MURDOCK / AEP MEDFORD 100.0 kV L	Reconductor line	\$482,000	58% (\$281,000)
NAAVINE / MURDOCK 100.0 kV	Reconductor line	\$3,943,000	58% (\$2,297,000)
SOUTH HAVEN SUBSTATION / NAAVINE 100.0 kV Line	Reconductor line	\$1,816,000	58% (\$1,058,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CANAL ST / LOWER GANITE 100.0 kV	Reconductor line	\$1,037,000	58% (\$604,000)
AEP MEDFORD / FEDERALGEN TAP 100.0 kV	Reconductor line	\$1,209,000	58% (\$704,000)
	Total allocated to project		\$4,944,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.